Newton's Law of Cooling

Recall that Newton's Law of cooling says that the rate of change of the temperature of an object is proportional to the difference between the object's temperature (denoted $T(t)$)and the temperature of the surroundings (T^*).
Write this down as a differential equation.
In practice, we do not usually know the value of the constant k at the start of the problem. Instead, problems involving Newton's Law of Cooling are often expressed as *boundary value problems*. That is, we measure the temperature twice and use the second measurement to determine k.

1. □Regrade The temperature in a certain Duke dorm is a stuffy 76°F at midnight when the power goes out. Outside the temperature holds steady at 35°F. The students measure the temperature in the dorm at 1 am and find that it has decreased to 65°F. Worried, they compute how low the temperature will fall by 7 am when Duke Energy Corporation has pledged that the heat will come back on. Recreate their calculation.

Often, loosening some our assumptions leads to much harder differential equations than the ones we saw on the previous worksheet. Below, we will see an examples of this.

Air Conditioning

Suppose we have a coffee cup whose temperature is 110°F. This time, though, instead of the room being at a fixed temperature, it is air conditioned. Its temperature varies sinusoidally between 67°F and 73°F with a period of 2 hours. Suppose that at time $t = 0$ the temperature of the room is 70°F (note: that is the midline!).

The ambient temperature T^* is a function of time instead of being constant. Explain why

$$T^*(t) = 3\sin(\pi t) + 70 \text{ degrees F}.$$

Refer back to the Modeling with Trig Functions lab!

Assuming that $k = 1$, write down an initial value problem for this situation.

2. Your equation is not separable. Use Euler's method in a spreadsheet to model the solution. Take $\Delta t = 0.1$ hours and compute data points for 15 hours. Plot your solution and sketch (or better: paste in) a graph below.

□Regrade

3. After a while, the temperature settles into what appears to be a sinusoidal pattern
□Regrade with period 2 hours. Set your y-axis to be between 67°F and 73°F. Sketch (or better: paste in) a graph below. You should see that the amplitude of the oscillation is not 3°F, like the amplitude of the temperature in the room. What is it, approximately? Explain why the temperature of the coffee doesn't vary as much as the temperature of the room.

4. Now change your k to be 4 instead of 1. Redo the graphs above in your sheet, and copy
□Regrade or paste in the second here. Approximately what is the amplitude of the long-term oscillation of the temperature of the cup? Does the cup's temperature settle into the oscillation faster or slower? In terms of what k means, explain.

Mixing Problems

Recall: if we have a container into which we add a substance at a certain rate and remove the substance at another rate, then the rate of change of the amount of the substance in the container is given by the difference between the two rates. That is:

$$\text{rate of change} = \text{rate in} - \text{rate out}$$

Suppose that we have a tank of salt water has 100 pounds of salt dissolved in 1000 gallons of water, except that unlike on the previous sheet 2 pounds of salt *but no water* is added per minute. As before, 4 gallons of brine are pumped out each minute.

5. Let $S(t)$ be the amount of salt in the container at time t minutes.

□Regrade

(a) What will be the volume of liquid in the tank t minutes after the start? What will be the concentration of salt in the tank (in terms of S and t)? Include units.

Volume: _____ Concentration: _____

(b) How much salt is being added each minute? How much is being removed? Include units. (One of these two answers will contain t and S.)

Added: _____ Removed: _____

(c) Write down (but do not try to solve) an initial value problem for $S(t)$.

$$\frac{dS}{dt} = _____, \quad S(0) = _____$$

(d) How long can this situation continue for? (Hint: See part (a)!)

(e) Does this equation have any equilibria? Show all your work.

6. Use Euler's method in a spreadsheet to compute of $S(t)$. Use $\Delta t = 0.1$. Model your □Regrade solution until the maximum time you found above. Sketch a graph of $S(t)$ below.

(a) You should see that the amount of salt in the tank rises for a while, then sharply falls. Look at your data and write down the maximum amount of salt in the tank to the nearest 0.1 lb, and the time at which it occurs to the nearest 0.1 minute.

Max amount: _____ at _____.

(b) After this time, the amount of salt in the tank decreases. How can that be the case if we continue to add 2 lbs of salt each minute?

7. At 77°F, the solubility of salt in water is about 3 lbs/gal. Add a column to your □Regrade spreadsheet computing the concentration of salt in the water. At what point in time does the salt no longer dissolve, and starts accumulating in a solid pile in the container? Write your answer to the nearest 0.1 minute.